NICHOLAS RUFFIN

Manager, Data Engineering · Commercial Pharmacy Data & RWE
Ellisville, MO | nruffinsecure@gmail.com | linkedin.com/in/nicholas-ruffin

PROFESSIONAL SUMMARY

I am a program leader in the retail pharmacy space with an engineering background, building the platforms and running the operations behind multiple product lines. I own commercial data delivery for a pharmacy-data monetization program built on NCPDP pharmacy transactions, and I manage the engineering team that builds and runs it. We deliver across three motions: de-identified patient-level tokenized claims, identified PHI delivered into customer-controlled AWS or Azure environments, and cross-source combined feeds joining multiple pharmacy systems. Tokenization is what makes the data longitudinal, and I own that de-identification lifecycle end to end, through Expert Determination certification and delivery into the customer's own environment. I also manage the delivery of reporting back to drug manufacturers for our copay assistance, voucher, and clinical programs, where I own the engineering and the delivery commitments rather than the programs themselves. Before RedSail I was Director of Full Stack Development at Nimble Solutions, where I built the accounts receivable platform that replaced spreadsheet collections with revenue-ordered work, running the software, RPA, and data teams that built the application and supplied the data behind it.

PROGRAMS

Data Monetization	A de-identified dataset built on unmodified, native NCPDP transmissions captured at the point of dispense, as the pharmacy sent them. That is what cannot be recreated: datasets assembled downstream from switches or payer adjudication are already modified and never see cash, discount-card, rejected, or reversed activity, and the source is an installed pharmacy base a competitor would have to own the software to reach. Serves patient-journey and real-world evidence platforms, market access and payer intelligence, life-sciences market and competitive intelligence, prescriber-level commercial analytics, financial and alt-data research, and community-pharmacy clinical networks. Tokenization is what unlocks it, letting customers stitch a patient journey across pharmacy systems and link it to the medical claims and lab data they already hold.
Copay Assistance & Voucher	Exists to stop patients from abandoning a prescription at the counter because the copay is too high. The pharmacy keeps the fill and the revenue instead of returning it to stock; the manufacturer gets a patient who starts and stays on therapy, plus the claim-level visibility to prove the program worked; we bill on the program activity we process and deliver the reporting back that the manufacturer measures performance and satisfies contractual audit obligations with. I own the engineering and the delivery commitments, not the program.
Clinical	Exists to change a filling behaviour at the counter: a lapsed patient filling again, a 30-day patient moved to 90-day, a formulary switch the plan now prefers. We find those patients in our own transaction data, turn each into a care goal, and put it in front of pharmacy staff inside the dispensing software, which is the only place a patient who is not filling can reliably be reached. The pharmacy earns on each goal its staff engages, the manufacturer gets the behaviour changed plus the outcome reporting that wins the renewal, and we bill on engagement and report on outcome. Same boundary: engineering and delivery commitments owned, program direction owned by the program team.

EXPERIENCE

Manager, Data Engineering | RedSail Technologies

Feb 2025 – Present

Data Monetization program leader within the RedSail Advantage Solutions team. I own engineering and delivery for the monetization program and its customer feeds, and deliver the data feeds behind the copay assistance, voucher, and clinical programs owned by adjacent program teams.

- **Data Monetization Program** Our software sits directly inside the pharmacy and our switch carries the claim, so we capture the raw NCPDP transmission whole: the request the pharmacy sent, the routing header from the switch, and the response the processor returned, whether that processor is a PBM, a cash program, or a discount card. Prescriber, patient, drug, quantity, days supply, and fill detail arrive alongside the adjudication itself, including what the plan paid and what the patient was charged at the counter, which is a record no one downstream of us can reconstruct. I shaped that raw feed into one coherent, readable, rights-checked data product, and because the same asset answers a different question for each buyer, it sells many times over instead of being rebuilt per deal: therapy history for real-world evidence and patient-journey platforms, coverage and copay reality for market access, dispensing behaviour for prescriber-level commercial analytics, and demand signal for financial and alt-data research. The program is now a $10M+ revenue line across 12 enterprise customers, licensed on multi-year agreements that have renewed and expanded every cycle.
  - **Patient-level tokenization.** A customer evaluating the dataset wants to know how much of their own population we actually cover before they buy, and afterward wants our dispensing history attached to the patients they already track. Patient-level tokenization is what makes that possible: it unlocks patient matching through overlap analysis against the customer's own data, then longitudinal linkage into the medical claims and lab data they already hold. That is what opened the RWE aggregator and patient-journey segment to us at all, because without a token our data is a snapshot of a fill, and with one it becomes a therapy history they can follow across pharmacies and link to the rest of a patient's care.
  - **Expert Determination.** Nothing could be sold commercially without it. Expert Determination, paired with the data-rights terms in our pharmacy agreements, is what ungates commercial use of the data, and it is the evidence that carries a customer's legal review. I run that process end to end with third-party evaluators, including re-certification as fields and populations change, across a network of 18,000+ active pharmacies with roughly 17,000 rights-enabled for commercial use.

- **Platform consolidation after the BestRx and PrimeRx acquisitions.** Each acquisition brought its own pharmacy system and its own claim shape, which meant more source data but also every existing customer feed suddenly sitting on an inconsistent foundation. I brought all of those data feeds under one program and drove consolidation onto a shared combined-feed pipeline, which meant meeting with the acquired leadership to agree the new data shape, then taking that shape back through renewal reviews with existing customers so each one signed off on what was changing and what their delivery would look like after. Per-customer tokenization and processing cost came down, redundant legacy pipelines retired, and no live customer lost delivery continuity through the cutover.
- **Delivery and client management.** The failure mode in data licensing is a contract signed on terms engineering cannot hold, discovered after the first delivery. I run delivery across a multi-cloud estate, into customer-controlled AWS and Azure environments for the customers who run their own tokenization engines and will not take a vendor's token, and over SFTP and cloud storage for the rest, with the access model, encryption, and proof-of-delivery evidence the contracts require. I carry the client relationships through onboarding, expansion, and renewal, and contracts come to me for review alongside business development and sales, where I hold the line on delivery methods, service levels, field-level agreement, and fill rates. The coverage volumes and sample files prospects evaluate before they buy come from me, as does the pricing of historical backfills as line items on dataset sales.
- **Team.** I manage a team of data engineers, assigning end-to-end ownership of customer-facing feed migrations and cutovers by account rather than by task, so each engineer carries a customer and I carry the program. Code standards and deployment discipline across the team remove single-person dependencies from customer delivery, which is what lets a small team run a portfolio this size.

- **RxAdvantage (Copay Assistance and Voucher Reporting)** These are owned by adjacent program teams; I own the engineering and the delivery commitments behind them. Manufacturers funding the programs cannot see what their money bought without claim-level reporting back from the pharmacies running them, and it reaches the pharmacy as part of what our software already does rather than as another line they have to buy, which is why the programs scale across the install base at all.
  - **Copay assistance and voucher reporting.** Shows the manufacturer how much abandonment at the counter the program prevented: the fills that happened because the patient's cost came down, the share the program absorbed, and the audit evidence the contract requires.
  - **Clinical program reporting.** Shows whether the behaviour the program paid for actually changed, measured on the same patients before and after, which is what the manufacturer renews on.
  - **Operational delivery motion.** DUA-governed and held to latency, coverage, and reliability rather than the analytics motion of the monetization platform, a different engineering problem and a different set of commitments.

Stack: Snowflake (multi-account, Procs/Streams/Tasks) · AWS (IAM, Lambda, S3, EventBridge, MWAA, Fargate, ECS) · Azure · Databricks and Iceberg tables (PMS team sources) · Apache Airflow · Terraform · GitHub Actions · SFTP · Python · SQL

Director, Full Stack Development | Nimble Solutions

Aug 2021 – Feb 2025

Software engineering and RPA leader under the IT organization, across onshore and offshore operations, through the company's inorganic growth phase.

- **Accounts Receivable Application Platform** Before it existed, collectors worked unpaid claims out of spreadsheets at roughly 20 touches a day, with no way to tell which claims were worth the call, so the highest-recovery claims aged alongside the worthless ones. I owned the application and the team that built it, and we replaced the spreadsheets with a work queue that ordered every rep's day by recoverable revenue backed by historical AR aging. Throughput went past 30 claims a day per rep, and the work that got done first was the work worth doing.
  - **Cloud function apps and custom API integrations.** The queue was only as good as the claim data behind it, and that data lived in practice management systems and clearinghouses, some with an interface and most without. The software team built Azure function apps on schedules plus custom API integrations through Availity and Waystar, pulling claim status, remittance and denial detail, and eligibility. That is what gave the application a current picture of where every claim stood, instead of a stale extract someone assembled by hand.
  - **RPA integration feeding the warehouse.** For the systems with no interface at all, the only data that existed was a report a human could run. The RPA team automated those report pulls and landed them in the data lake, where the data team normalized them through ETL into the warehouse the application consumed. That path is what made each newly acquired practice management system a configuration problem instead of a rebuild.
  - **Acquisition integration.** Four healthcare acquisitions in the growth phase, each arriving with its own stack and its own systems to absorb. Technical partner to the CIO on due diligence, TCO analysis, integration sequencing, and stack rationalization, then integrated the acquired applications into our platform; traveled to Hyderabad to fold the acquired team into US operations.
  - **Team.** Built out and managed 16+ engineers across three functions, software, RPA, and later data, split between the US and Hyderabad, with a project manager coordinating delivery across them so I could stay on architecture and the CIO's priorities rather than on scheduling. Hired into both teams and set the CI/CD and test-coverage standards they delivered against.

Stack: Azure (Function Apps, Active Directory for RBAC, Application Insights for monitoring and observability, Blob Storage, SQL Server) · C#/.NET microservices · RPA automation · ETL into a unified data warehouse · CI/CD

Senior Full Stack Developer | Core & Main

Jun 2017 – Aug 2021

- Developed enterprise data applications using C#/.NET and SQL, supporting 200+ concurrent corporate users across operations.

EDUCATION

Bachelor of Science, Computer Science — University of Phoenix